

8. A satellite of mass M is in circular orbit with radius r around the Earth with a period of T . A second satellite of mass $2M$ moves around the Earth with radius $4r$. What is the period of this second satellite in terms of T ?

A $\frac{T}{4}$

B $2T$

C $4T$

D $8T$

Ans: D

For the 1st satellite,

$$\frac{GM_E M}{r^2} = Mr\omega^2$$

$$\frac{GM_E M}{r^2} = Mr\left(\frac{2\pi}{T}\right)^2$$

$$T = 2\pi\sqrt{\frac{r^3}{GM_E}}$$

$$\text{For 2nd satellite, } T' = 2\pi\sqrt{\frac{(4r)^3}{GM_E}} = 8T$$

8

9 What is the approximate number of atoms in a cubic metre of an ideal monatomic gas